Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)
Chapter 3: Guidelines, Principles, and Theories

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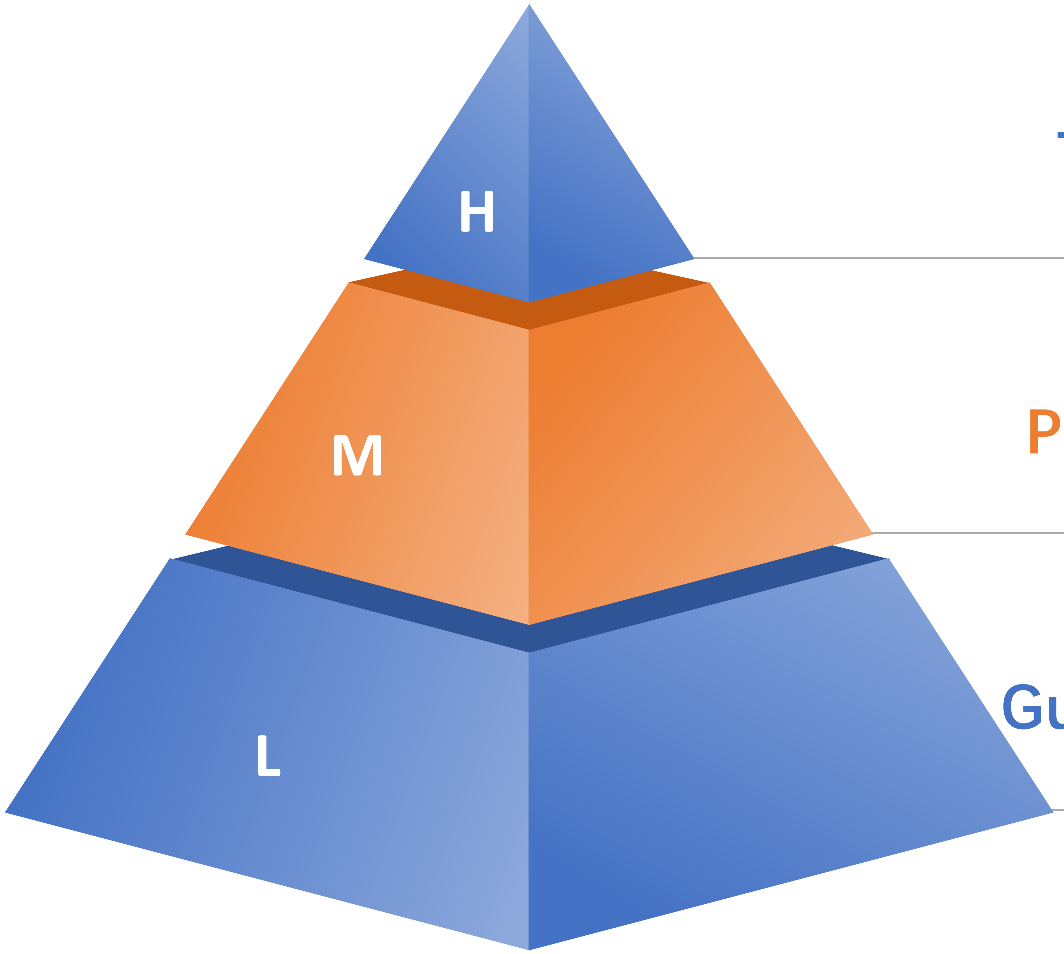
Introduction

- Three-Level Organization
  - UIDers have accumulated a wealth of experience and researchers have produced a growing body of empirical evidence and theories, all of which can be organized into three levels of knowledge
  - Guidelines
    - Low-level focused advice about good practices and cautions against dangers
  - Principles
    - Middle-level strategies or rules to analyse and compare UID alternatives
  - Theories
    - High-level widely applicable frameworks to draw on during UID/evaluation to support communication and teaching
    - Theories can also be predictive

Introduction

- Three-Level Organization



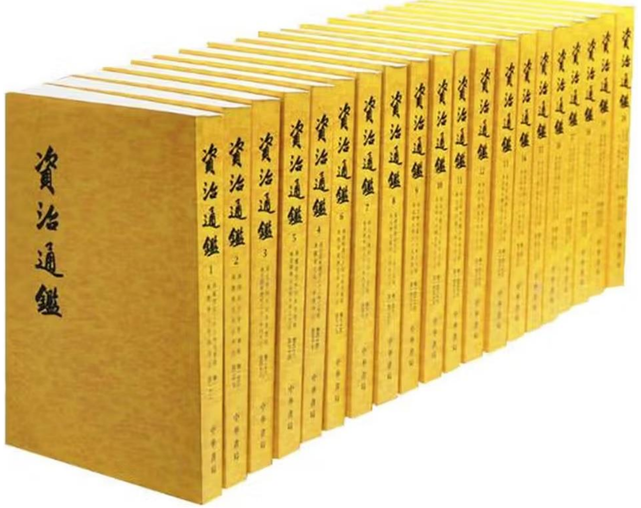
Theories High-level widely applicable frameworks

Principles Middle-level strategies or rules

Guidelines Low-level focused advice

Guidelines

- Basics
  - From the earliest days of computing, UIDers have written down guidelines to record their insights and try to guide the efforts of future UIDers
  - A guidelines document helps by developing a shared language and then promoting consistency among multiple UIDers in terminology usage, appearance, and action sequences
  - A guidelines document records best practices derived from practical experience or empirical studies, with appropriate examples/counterexamples



围魏救赵	围点打援	纸上谈兵	朝秦暮楚
春秋笔法	折冲樽俎	犯汉必诛	唇亡齿寒
秦晋之好	一飞冲天	问鼎中原	城下之盟
百步穿杨	狡兔三窟	纵横捭阖	卧薪尝胆
三顾茅庐	鸿鹄之志	经天纬地	入乡问俗
萧规曹随	国士无双	好整以暇	变法图强

Guidelines

- Navigating the UI
  - Plenty of guidelines have been accumulated to promote the design of informative webpages and navigation-oriented UIs
  - Standardize task sequences
    - Allow users to perform tasks in the same sequence and manner across similar conditions
  - Ensure that embedded links are descriptive
    - When using links, the link text should accurately describe the link’s destination
  - Use unique and descriptive headings
    - Use headings that are distinct from one another and conceptually related to the content
  - Use radio buttons for mutually exclusive choices
    - Provide a radio button control when a list of mutually exclusive options is involved
  - Develop pages that will print properly
    - If users are likely to print one or more pages, develop pages with widths that print properly

Guidelines

- Navigating the UI
  - By relative importance

Guideline Heading	Relative Importance
Provide Useful Content	5
Establish User Requirements	5
Understand and Meet User's Expectations	5
Involve Users in Establishing User Requirements	5
Do Not Display Unsolicited Windows or Graphics	5
Comply with Section 508	5
Design Forms for Users Using Assistive Technology	5
Do Not Use Color Alone to Convey Information	5
Enable Access to the Homepage	5
Show All Major Options on the Homepage	5
Create a Positive First Impression of Your Site	5
Avoid Cluttered Displays	5
Place Important Items Consistently	5
Place Important Items at Top Center	5
Eliminate Horizontal Scrolling	5
Use Clear Category Labels	5
Use Meaningful Link Labels	5
Distinguish Required and Optional Data Entry Fields	5
Label Pushbuttons Clearly	5
Make Action Sequences Clear	5
Organize Information Clearly	5
Facilitate Scanning	5
Ensure that Necessary Information is Displayed	5
Ensure Usable Search Results	5
Design Search Engines to Search the Entire Site	5
Set and State Goals	4
Focus on Performance Before Preference	4
Consider Many User Interface Issues	4
Be Easily Found in the Top 30	4
Increase Web Site Credibility	4
Standardize Task Sequences	4
Reduce the User's Workload	4
Design For Working Memory Limitations	4
Minimize Page Download Time	4
Warn of 'Time Outs'	4
Display Information in a Directly Usable Format	4
Format Information for Reading and Printing	4
Provide Feedback when Users Must Wait	4
Inform Users of Long Download Times	4
Develop Pages that Will Print Properly	4
Enable Users to Skip Repetitive Navigation Links	4
Provide Text Equivalents for Non-Text Elements	4

Guideline Heading	Relative Importance
Test Plug-Ins and Applets for Accessibility	4
Design for Common Browsers	4
Account for Browser Differences	4
Design for Popular Operating Systems	4
Design for User's Typical Connection Speed	4
Communicate the Web Site's Value and Purpose	4
Limit Prose Text on the Homepage	4
Ensure the Homepage Looks like a Homepage	4
Structure for Easy Comparison	4
Establish Level of Importance	4
Optimize Display Density	4
Align Items on a Page	4
Provide Navigational Options	4
Differentiate and Group Navigation Elements	4
Use a Clickable 'List of Contents' on Long Pages	4
Provide Feedback on Users' Location	4
Place Primary Navigation Menus in the Left Panel	4
Provide Descriptive Page Titles	4
Use Descriptive Headings Liberally	4
Use Unique and Descriptive Headings	4
Highlight Critical Data	4
Use Descriptive Row and Column Headings	4
Link to Related Content	4
Match Link Names with Their Destination Pages	4
Avoid Misleading Cues to Click	4
Repeat Important Links	4
Use Text for Links	4
Designate Used Links	4
Use Black Text on Plain, High-Contrast Backgrounds	4
Format Common Items Consistently	4
Use Mixed-Case for Prose Text	4
Ensure Visual Consistency	4
Order Elements to Maximize User Performance	4
Place Important Items at Top of the List	4
Format Lists to Ease Scanning	4
Display Related Items in Lists	4
Label Data Entry Fields Consistently	4
Do Not Make User-Entered Codes Case Sensitive	4
Label Data Entry Fields Clearly	4
Minimize User Data Entry	4
Use Simple Background Images	4

Guidelines

- Navigating the UI
  - By relative importance

Guideline Heading	Relative Importance
Ensure that Images Do Not Slow Downloads	4
Use Video, Animation, and Audio Meaningfully	4
Include Logos	4
Graphics Should Not Look like Banner Ads	4
Limit Large Images Above the Fold	4
Ensure Web Site Images Convey Intended Messages	4
Avoid Jargon	4
Use Familiar Words	4
Define Acronyms and Abbreviations	4
Use Abbreviations Sparingly	4
Use Mixed Case with Prose	4
Limit the Number of Words and Sentences	4
Group Related Elements	4
Minimize the Number of Clicks or Pages	4
Make Upper- and Lowercase Search Terms Equivalent	4
Provide a Search Option on Each Page	4
Design Search Around Users' Terms	4
Use an Iterative Design Approach	4
Set Usability Goals	3
Do Not Require Users to Multitask While Reading	3
Use Users' Terminology in Help Documentation	3
Provide Printing Options	3
Ensure that Scripts Allow Accessibility	3
Provide Equivalent Pages	3
Provide Client-Side Image Maps	3
Synchronize Multimedia Elements	3
Do Not Require Style Sheets	3
Design for Commonly Used Screen Resolutions	3
Limit Homepage Length	3
Use Fluid Layouts	3
Avoid Scroll Stoppers	3
Set Appropriate Page Lengths	3
Use Moderate White Space	3
Use Descriptive Tab Labels	3
Present Tabs Effectively	3
Use Headings in the Appropriate HTML Order	3
Provide Consistent Clickability Cues	3
Ensure that Embedded Links are Descriptive	3
Use 'Pointing-and-Clicking'	3
Use Appropriate Text Link Lengths	3
Indicate Internal vs. External Links	3

Guideline Heading	Relative Importance
Link to Supportive Information	3
Use Bold Text Sparingly	3
Use Attention-Attracting Features when Appropriate	3
Use Familiar Fonts	3
Use at Least a 12-Point Font	3
Introduce Each List	3
Use Static Menus	3
Put Labels Close to Data Entry Fields	3
Allow Users to See Their Entered Data	3
Use Radio Buttons for Mutually Exclusive Selections	3
Use Familiar Widgets	3
Anticipate Typical User Errors	3
Partition Long Data Items	3
Use a Single Data Entry Method	3
Prioritize Pushbuttons	3
Use Check Boxes to Enable Multiple Selections	3
Label Units of Measurement	3
Do Not Limit Viewable List Box Options	3
Display Default Values	3
Limit the Use of Images	3
Include Actual Data with Data Graphics	3
Display Monitoring Information Graphically	3
Limit Prose Text on Navigation pages	3
Use Active Voice	3
Write Instructions in the Affirmative	3
Make First Sentences Descriptive	3
Design Quantitative Content for Quick Understanding	3
Display Only Necessary Information	3
Format Information for Multiple Audiences	3
Allow Simple Searches	3
Notify Users when Multiple Search Options Exist	3
Include Hints to Improve Search Performance	3
Solicit Test Participants' Comments	3
Evaluate Web Sites Before and After Making Changes	3
Prioritize Tasks	3
Distinguish Between Frequency and Severity	3
Select the Right Number of Participants	3
Use Parallel Design	2
Provide Assistance to Users	2
Provide Frame Titles	2
Avoid Screen Flicker	2

Guidelines

- Navigating the UI
  - By relative importance
  - By strength of evidence

Guideline Heading	Relative Importance
Attend to Homepage Panel Width	2
Choose Appropriate Line Lengths	2
Keep Navigation-Only Pages Short	2
Use Appropriate Menu Types	2
Use Site Maps	2
Facilitate Rapid Scrolling While Reading	2
Use Scrolling Pages for Reading Comprehension	2
Use Paging Rather Than Scrolling	2
Scroll Fewer Screenfuls	2
Provide Users with Good Ways to Reduce Options	2
Color-Coding and Instructions	2
Emphasize Importance	2
Highlighting Information	2
Start Numbered Items at One	2
Use Appropriate List Style	2
Place Cursor in First Data Entry Field	2
Ensure that Double-Clicking Will Not Cause Problems	2
Use Open Lists to Select One from Many	2
Use Data Entry Fields to Speed Performance	2
Use a Minimum of Two Radio Buttons	2
Provide Auto-Tabbing Functionality	2
Introduce Animation	2
Emulate Real-World Objects	2
Use Thumbnail Images to Preview Larger Images	2
Use Color for Grouping	2
Provide Search Templates	2
Use the Appropriate Prototyping Technology	2
Use Inspection Evaluation Results Cautiously	2
Recognize the 'Evaluator Effect'	2
Use Personas	1
Use Frames When Functions Must Remain Accessible	1
Use 'Glosses' to Assist Navigation	1
Breadcrumb Navigation	1
Capitalize First Letter of First Word in Lists	1
Minimize Use of the Shift Key	1
Use Images to Facilitate Learning	1
Using Photographs of People	1
Apply Automatic Evaluation Methods	1
Use Cognitive Walkthroughs Cautiously	1
Choosing Laboratory vs. Remote Testing	1
Use Severity Ratings Cautiously	1

Guideline Heading	Strength of Evidence
Provide Useful Content	5
Standardize Task Sequences	5
Design for Working Memory Limitations	5
Align Items on a Page	5
Use Descriptive Headings Liberally	5
Use Black Text on Plain, High-Contrast Backgrounds	5
Use Attention-Attracting Features when Appropriate	5
Use Familiar Fonts	5
Emphasize Importance	5
Order Elements to Maximize User Performance	5
Use Data Entry Fields to Speed Performance	5
Use Simple Background Images	5
Use Video, Animation, and Audio Meaningfully	5
Use Images to Facilitate Learning	5
Use Mixed Case with Prose	5
Group Related Elements	5
Use Color for Grouping	5
Use an Iterative Design Approach	5
Establish User Requirements	4
Be Easily Found in the Top 30	4
Use Parallel Design	4
Minimize Page Download Time	4
Provide Feedback When Users Must Wait	4
Do Not Require Users to Multitask While Reading	4
Do Not Use Color Alone to Convey Information	4
Create a Positive First Impression of Your Site	4
Ensure the Homepage Looks like a Homepage	4
Place Important Items Consistently	4
Place Important Items at Top Center	4
Structure for Easy Comparison	4
Avoid Scroll Stoppers	4
Use Moderate White Space	4
Choose Appropriate Line Lengths	4
Use Frames when Functions Must Remain Accessible	4
Keep Navigation-Only Pages Short	4
Use Appropriate Menu Types	4
Use Site Maps	4
Eliminate Horizontal Scrolling	4
Facilitate Rapid Scrolling While Reading	4
Use Scrolling Pages for Reading Comprehension	4
Use Paging Rather Than Scrolling	4
Use Clear Category Labels	4

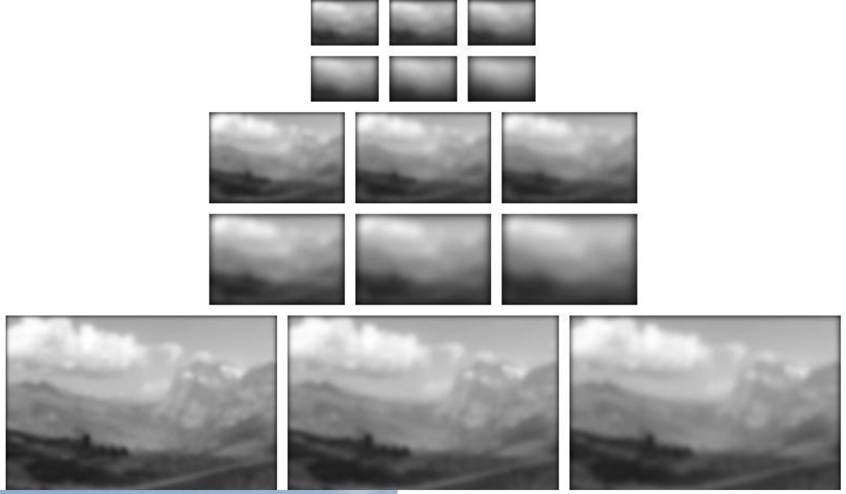
Guidelines

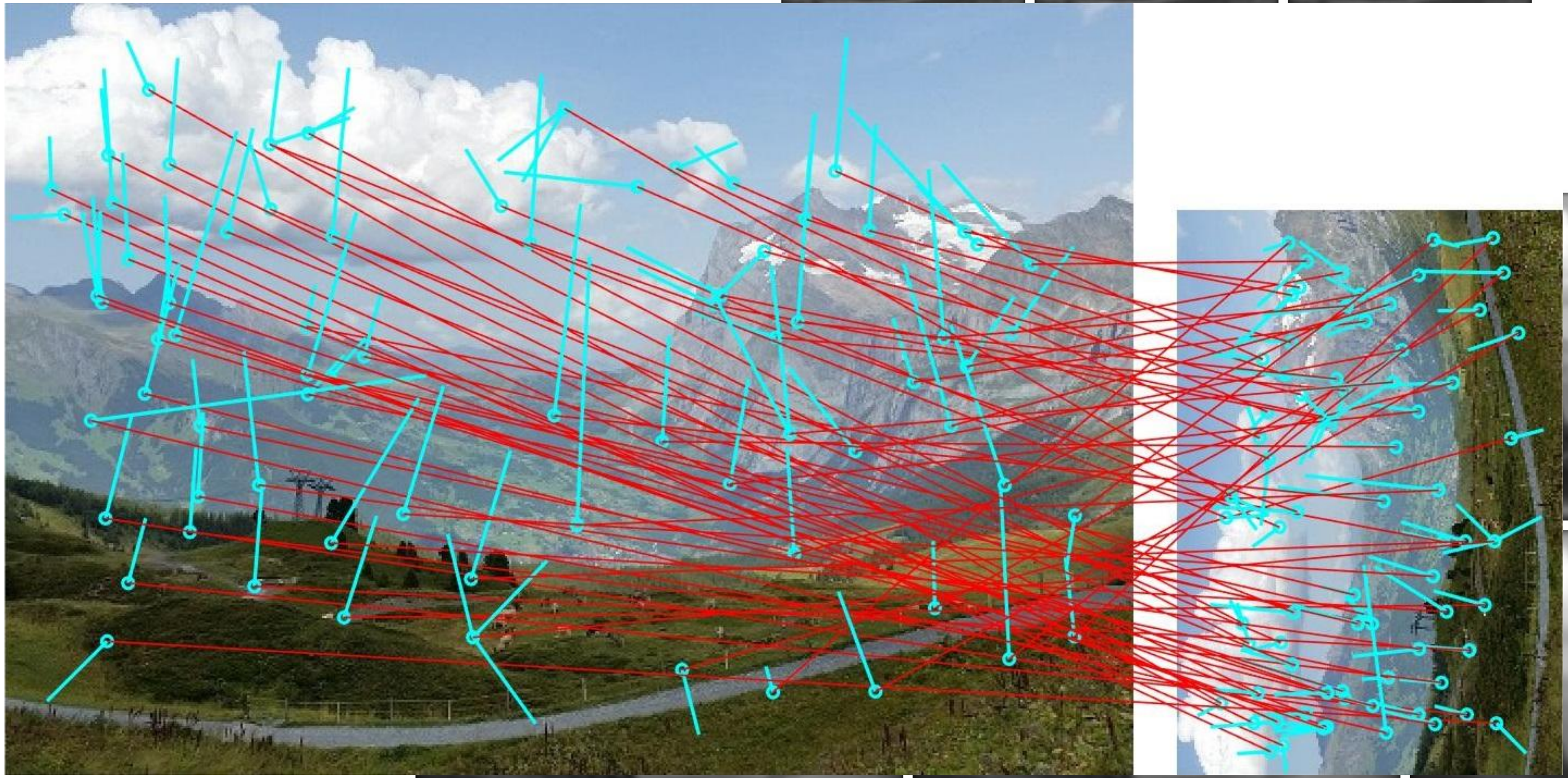
- Navigating the UI
  - Plenty of guidelines have been accumulated to promote the design of informative webpages and navigation-oriented UIs
  - Standardize task sequences
  - Ensure that embedded links are descriptive
  - Use unique and descriptive headings
  - Use radio buttons for mutually exclusive choices
  - Develop pages that will print properly
  - Use thumbnail images to preview larger images
    - When viewing full-size images is not critical, first provide a thumbnail of the image
    - Not purely empirical knowledge without knowing why, but biologically explainable

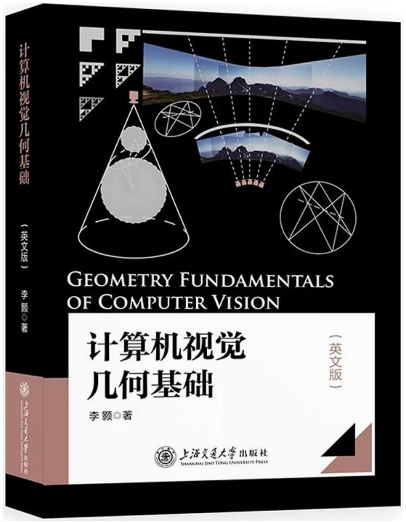
Scale-space theory ⟷ Biologically inspired/supported

Guidelines

- Navigating the UI
  - Use thumbnail images to preview



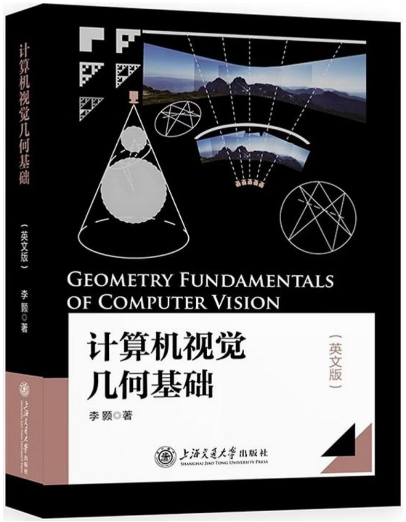


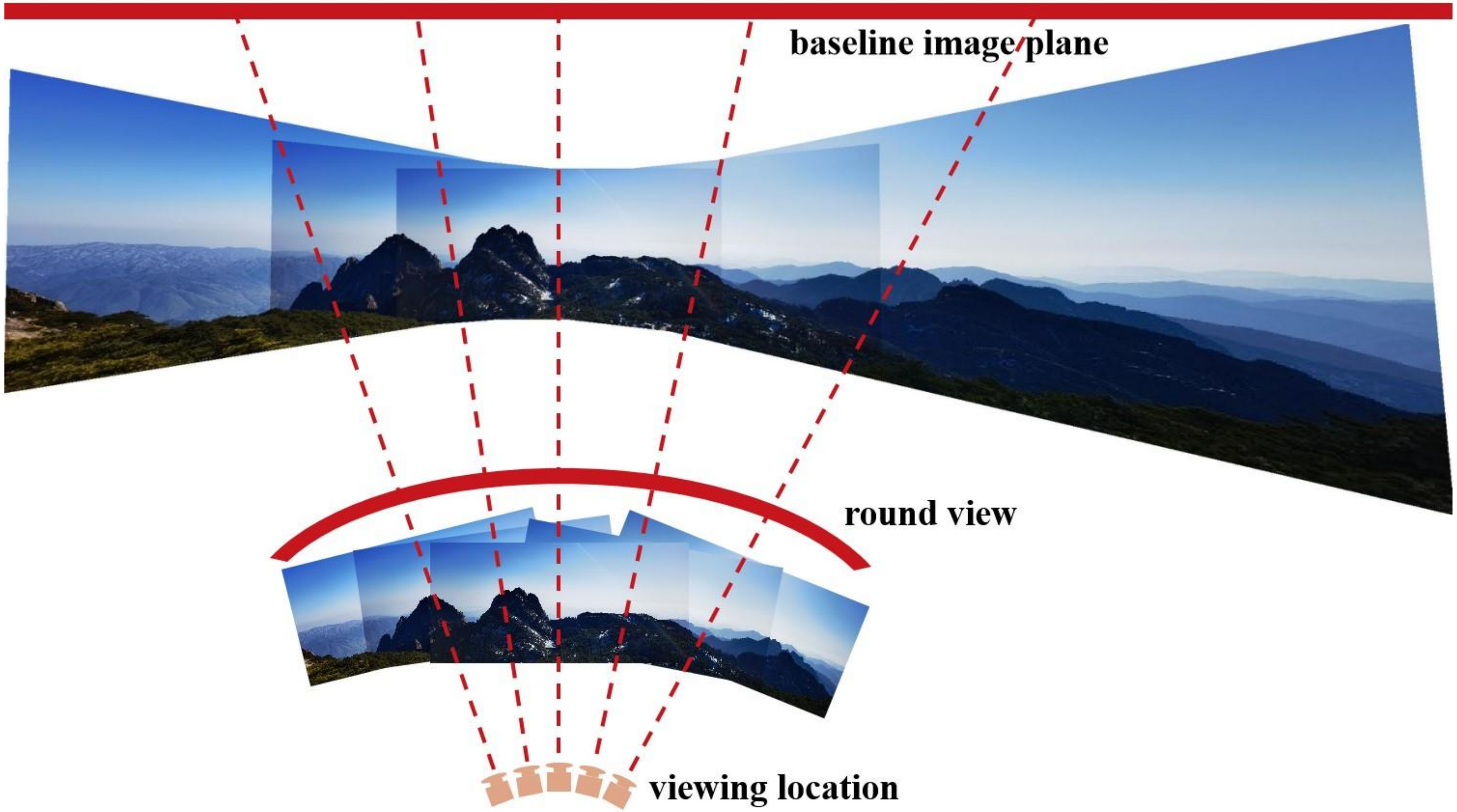


Scale-space theory

Guidelines

- Navigating the UI
  - Use thumbnail images to preview





Guidelines

- Navigating the UI
- Organizing the Display
    - Display development/design is a large topic with many special cases
    - Five high-level goals for data display
        - Consistency of data display
        - Efficient information assimilation by the user
        - Minimal memory load on the user
        - Compatibility of data display with data entry
        - Flexibility for user control of data display

Guidelines

- Organizing the Display
  - Consistency of data display
    - During the UID process, the terminology, abbreviations, formats, colours, capitalization, and so on should all be standardized and controlled by use of a dictionary of the items
  - Efficient information assimilation by the user
    - The format should be familiar to the operator and should be related to the tasks required to be performed with the data
    - This goal is served by rules of neat columns of data, left/right justification for alphabetic data/integers respectively, lining up of decimal points, proper spacing, use of comprehensible labels, appropriate measurement units, numbers of decimal digits, etc.
  - Minimal memory load on the user
    - Users should not be required to remember information from one screen for use on another screen
    - Tasks should be arranged such that completion occurs with few actions, minimizing the chance of forgetting to perform a step

Guidelines

- Organizing the Display
  - Consistency of data display
  - Efficient information assimilation by the user
  - Minimal memory load on the user
  - Compatibility of data display with data entry
    - The format of displayed information should be linked clearly to the format of data entry
    - The output fields, where possible and appropriate, should also act as editable input fields
  - Flexibility for user control of data display
    - Users should be able to get the information from the display in the form most convenient for the task on which they are working --- E.g. the order of columns nd sorting of rows should be easily changeable by the users

Guidelines

- Organizing the Display
  - Display development/design is a large topic with many special cases
  - Five high-level goals for data display
    - Consistency of data display
    - Efficient information assimilation by the user
    - Minimal memory load on the user
    - Compatibility of data display with data entry
    - Flexibility for user control of data display
  - Above compact set of high-level goals is a useful starting point, but each project needs to expand these into application-specific and hardware-dependent standards and practices

Guidelines

- Navigating the UI
- Organizing the Display
- Getting the User's Attention
  - Intensity
    - Use two levels only, with limited use of high intensity to draw attention
  - Marking
    - Underline the item, enclose it in box, point to it with an arrow, or use an indicator such as an asterisk, bullet, dash, plus sign, etc.
  - Size
    - Use up to four sizes, with larger sizes attracting more attention
  - Choice of fonts
    - Use up to three fonts

Guidelines

- Getting the User’s Attention
    - Intensity
    - Marking
    - Size
    - Choice of fonts
    - Blinking
        - Use blinking displays (2-4 Hz) or blinking colour changes with great care and in limited areas, as it is distracting and can trigger seizures
    - Colour
        - Use up to four standard colours, with additional colours reserved for occsional use
    - Audio
        - Use soft tones for regular positive feedback and harsh sound for rare emergency conditions

Guidelines

- Getting the User's Attention
  - Typical techniques
    - Intensity
    - Marking
    - Size
    - Choice of fonts
    - Blinking
    - Colour
    - Audio
  - Do not abuse any attention-getting technique
    - "Everything is important" = "Nothing is important"

Guidelines

- Navigating the UI
- Organizing the Display
- Getting the User's Attention
- Facilitating Data Entry
    - Five high-level goals for data entry
        - Consistency of data-entry transactions
        - Minimal input actions by the user
        - Minimal memory load on the user
        - Compatibility of data entry with data display
        - Flexibility for user control of data entry

- Five high-level goals for data display
    - Consistency of data display
    - Efficient information assimilation by the user
    - Minimal memory load on the user
    - Compatibility of data display with data entry
    - Flexibility for user control of data display

DISCUSSION

Q & A